

EX1. Find Your own vector space

Classification Theorem if $\dim V = n$, then $V \cong \mathbb{F}^n$

EX2. Find your own linear maps between V and W

F.T.O.L.A = Fundamental Thm of L.A.:

$\dim V = n < +\infty$, $\dim W = m < +\infty$ Then $\text{Hom}(V, W) \cong \mathbb{F}^{m \times n} = \text{Hom}(\mathbb{F}^n, \mathbb{F}^m)$

$\text{Hom}(V, W)$	$\mathbb{F}^{m \times n}$
	$0_{m \times n}$
$\sigma: V_i \mapsto \sigma(V_i) = W_i$ $n \leq m$ $V \mapsto 0$ $n > m$ $V_{m+1} \mapsto 0$	$1_{m \times n}$
Essence: $\sigma: V \mapsto \sigma(V) = \sum x_i W_i$ $\sum x_i V_i \mapsto [W_1, W_2, \dots, W_n] X$	

Cor. Fix basis B_V resp B_W of V resp W $\forall \sigma \in \text{Hom}(V, W)$

Let $V = B_V X$ then $\sigma(V) = B_W (P_\sigma X)$

In other words, σ sends V with coordinate x to W with $P_\sigma x$

$\text{Hom}(V, W)$

$\mathbb{F}^{m \times n}$

① $\dim \text{Im } \Phi(A) = r$

① $\text{rank } A = r \iff \dim \text{Ker}(A) = n - r$

②

② $\iff \text{equivalent } B = P A Q$

③ injection

③ full column rank ($n \leq m$)

④ surjection

④ full row rank ($n \geq m$)

EX. $V = \{y \mid y'' + y' - 2y = 0\} \cong \mathbb{F}^2$

$= \{c_1 e^{-2x} + c_2 e^x \mid c_1, c_2 \in \mathbb{F}\}$

"+" : $c_1 e^{-2x} + c_2 e^x + c'_1 e^{-2x} + c'_2 e^x \longrightarrow \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} c'_1 \\ c'_2 \end{pmatrix}$
 $= (c_1 + c'_1) e^{-2x} + (c_2 + c'_2) e^x = \begin{pmatrix} c_1 + c'_1 \\ c_2 + c'_2 \end{pmatrix}$

1. $\text{Hom}(\mathbb{F}, \mathbb{F}^\infty)$

or $\text{Hom}_{\mathbb{F}}(\mathbb{F}, W) \cong W$

$\sigma \longmapsto \Phi(\sigma) = \sigma(1)$

injection : $\sigma \in \text{Ker } \Phi \iff \text{i.e. } \Phi(\sigma) = 0 \iff \text{i.e. } \sigma(1) = 0 \Rightarrow \sigma(x) = x \sigma(1) = 0$

surjection :

What is $\mathbb{F}^\infty$? $\mathbb{F}^\infty = \{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_i \in \mathbb{F} \}$

① $\mathbb{F}^\infty = \{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid \text{all but finitely many } x_i \text{ are zero} \}$

coproduct $\bigoplus_{i=1}^{\infty} \mathbb{F} = \text{direct sum} \cong \mathbb{F}[X] \quad \begin{pmatrix} a_0 \\ \vdots \\ a_n \end{pmatrix} \longmapsto \sum_{i=0}^n a_i X^i$

② $\mathbb{F}^\infty = \{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_i \in \mathbb{F} \}$

($\mathbb{F}^\infty$) product $\bigcap_{i=1}^{\infty} \mathbb{F} \cong \mathbb{F}[X]$ (幂级数) $\begin{pmatrix} a_0 \\ \vdots \\ a_n \end{pmatrix} \cong \sum_{i=0}^n a_i X^i$

2. $\text{Hom}(\bigoplus_{i=1}^{\infty} \mathbb{F}, \mathbb{F}) \cong \text{Hom}(\mathbb{F}[X], \mathbb{F}) \rightarrow \sigma$

$\sigma : 1, X, \dots, X^n, \dots \longmapsto \sigma(X^i) = \lambda_i$

$f(x) = \sum_{i=0}^n a_i X^i \longmapsto \sum_{i=0}^n \lambda_i a_i$

$\sigma \longmapsto \sum_{i=0}^{\infty} \lambda_i X^i \in \mathbb{F}[X]$

$\text{Hom}(\mathbb{F}[X], \mathbb{F}) \cong \mathbb{F}[X]$

$$\text{Ex. } \mathbb{R} \rightarrow \mathbb{R}^{>0}$$

$$x \mapsto e^x$$

want this to be linear
 $\Rightarrow (\mathbb{R}^{>0}, \oplus = \cdot, \otimes = +)$

$$x+y \mapsto e^{x+y} = e^x \cdot e^y$$

$$\sigma(x+y) = e^{x+y} = e^x \cdot e^y = \sigma(x) \otimes \sigma(y) = e^x \otimes e^y$$

$$\cdot (V_1 \times V_2) \times V_3 \cong V_1 \times (V_2 \times V_3)$$

$$((v_1, v_2), v_3) \mapsto (v_1, (v_2, v_3))$$

up to isomorphism, $V_1 \times V_2 \times \dots \times V_n$ is unique

$$\cdot \dim V = n < \infty$$

$$V \cong \mathbb{F}^n$$

$$\text{if } V_i = 1 \text{ then } V_1 \otimes V_2 \otimes \dots \otimes V_n \cong \mathbb{F} \otimes \dots \otimes \mathbb{F}$$

$$V \times W = (v, 0) \oplus (0, w)$$

$$[V \cong (v, 0) \quad W \cong (0, w)]$$